

ASSIGNMENT 5

Title : Embedded C program for LCD interfacing with
PIC 18FXXX microcontroller

Objective : To learn interfacing of LCD screen with
PIC 18FXXX microcontroller

Problem Statement : Write an embedded C program for LCD
interfacing with PIC 18FXXX

Input:

```
#include <xc.h>
#include<p18f4550.h>
#define LCD_EN LATAbits.LA1
#define LCD_RS LATAbits.LA0
#define LCDPORT LATB

void lcd_delay(unsigned int time){
    unsigned int j, i;
    for(i=0; i<time; i++){
        for(j=0; j<100; j++){
        }
    }
}

void SendInstruction(unsigned char command){
    LCD_RS=0;
```

```

    LCDPORT=command;
    LCD_EN=1;
    lcd_delay(10);
    LCD_EN=0;
    lcd_delay(10);
}

```

```

void SendData(unsigned char LcdData){
    LCD_RS=1;
    LCDPORT=LcdData;
    LCD_EN=1;
    lcd_delay(10);
    LCD_EN=0;
    lcd_delay(10);
}

```

```

void initLCD(void){
    ADCON1=0x0F;
    TRISB=0x00;
    TRISAbits.RA0=0;
    TRISAbits.RA1=0;
    SendInstruction(0x38);
    SendInstruction(0x06);
    SendInstruction(0x0C);
    SendInstruction(0x01);
    SendInstruction(0x80);
}

```

```

unsigned char* string1="MicroEmbedded";
unsigned char* string2="PIC Kit";

```

```

void main(void) {
    ADCON1=0x0F;
    TRISB=0x00;

```

```
TRISAbits.RA0=0;
TRISAbits.RA1=0;
SendInstruction(0x38);
SendInstruction(0x06);
SendInstruction(0x0C);
SendInstruction(0x01);
SendInstruction(0x80);

while(*string1){
    SendData(*string1);
    string1++;
}
SendInstruction(0xC0);
while(*string2){
    SendData(*string2);
    string2++;
}
while(1);
return;
}
```